4. Bayesian Networks: Representation

CSC3043S Reasoning in AI: Bayesian Networks

Dr Jan Buys

University of Cape Town

[Based on slides created by Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley. All materials available at http://ai.berkeley.edu.]



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Conditional Independence

- Reminder: X is **conditionally independent** of Y given Z:

$$\forall x, y, z : P(x|z, y) = P(x|z) \dashrightarrow X \perp\!\!\!\perp Y|Z$$

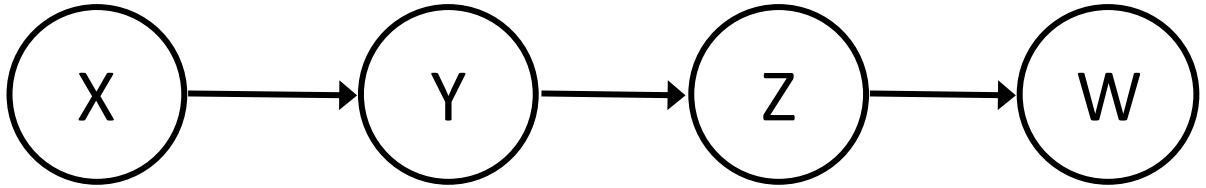
- Conditional independence is a property of a *distribution*
- We make some assumptions when defining a Bayesian network:

$$P(x_i|x_1 \cdots x_{i-1}) = P(x_i|parents(X_i))$$

- However, there may be *additional* conditional independencies in the distribution beyond these
- We will see how to read them off the graph

Conditional Independence

- Example:



- Joint probability definition compared to the chain rule:

$$P(X)P(Y|X)P(Z|X,Y)P(W|X,Y,Z)$$
$$= P(X)P(Y|X)P(Z|Y)P(W|Z)$$

- Can derive some conditional independence assumptions directly:

$$Z \perp\!\!\!\perp X|Y$$

$$W \perp\!\!\!\perp X,Y|Z$$

- Additional implied conditional independence assumption:

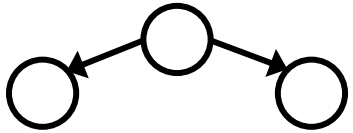
$$W \perp\!\!\!\perp X|Y$$

Conditional Independence

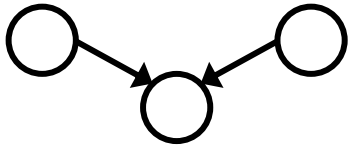
- **D-separation**: A procedure for deriving independencies from Bayesian network graphs
- Consider variable triples and determine whether there is (conditional) independence between then
- We will consider 3 cases from which any graph can be analyzed, then derive a general procedure



Causal chain



Common cause



Common effect

Conditional Independence

1. Causal chains



X: Low pressure Y: Rain Z: Traffic

$$P(x, y, z) = P(x)P(y|x)P(z|y)$$

- Cannot guarantee that X is independent of Z
- We can say that X is independent of Z **given** $\boldsymbol{Y}$:

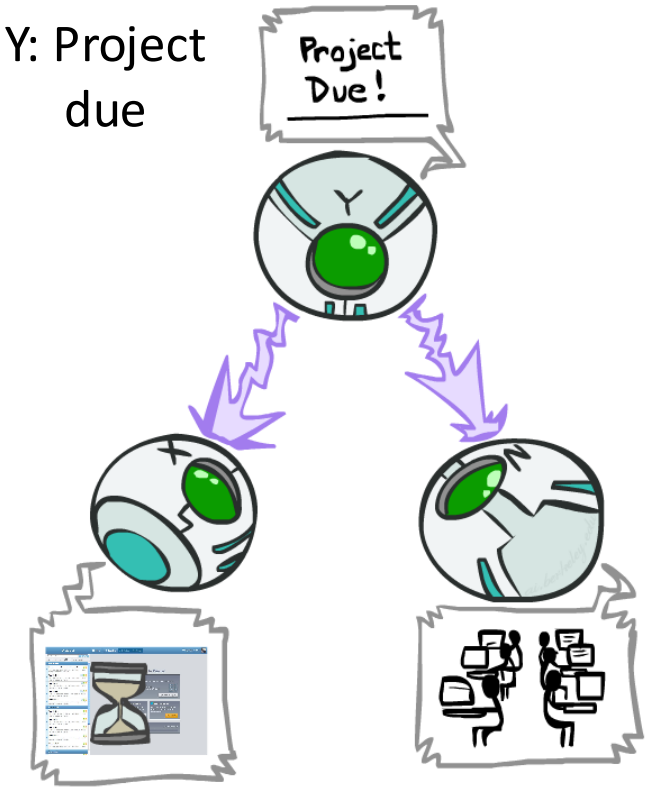
$$\begin{aligned} P(z|x, y) &= \frac{P(x, y, z)}{P(x, y)} \\ &= \frac{P(x)P(y|x)P(z|y)}{P(x)P(y|x)} \\ &= P(z|y) \end{aligned}$$

- Evidence along the chain "blocks" the influence

Conditional Independence

2. Common cause

Y: Project due



X: Forums busy

Z: Lab full

$$P(x, y, z) = P(y)P(x|y)P(z|y)$$

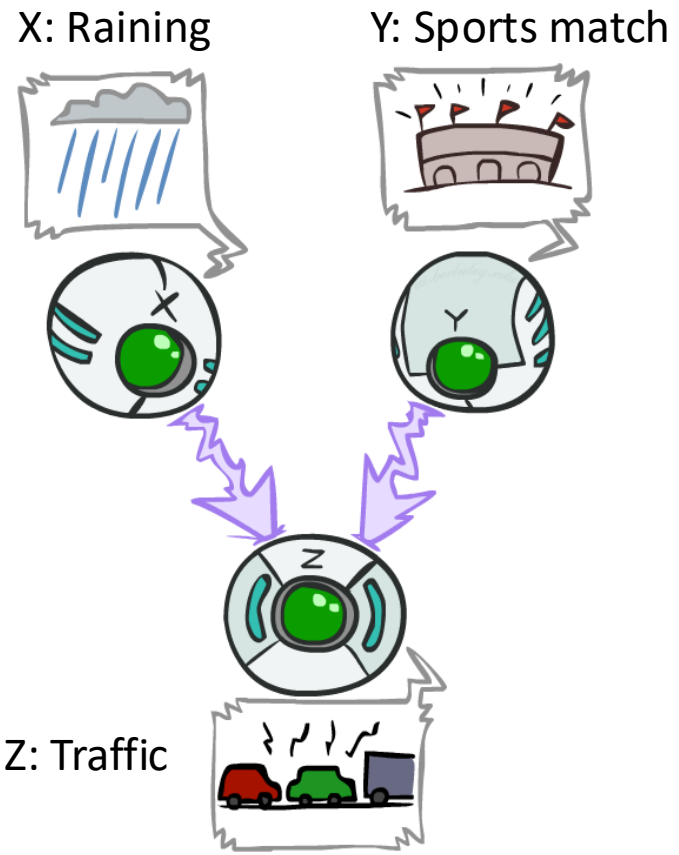
- Cannot guarantee that X is independent of Z
- Can say that X and Z are independent **given Y**

$$\begin{aligned} P(z|x,y) &= \frac{P(x,y,z)}{P(x,y)} \\ &= \frac{P(y)P(x|y)P(z|y)}{P(y)P(x|y)} \\ &= P(z|y) \end{aligned}$$

- Observing the cause blocks the influence between the effects

Conditional Independence

3. Common effect (v-structure)



$$P(x, y, z) = P(x)P(y)P(z|x, y)$$

- Here X **is** independent of Y

$$P(x|y) = P(x)$$

- X and Y are **not** conditionally independent given Z
- This is *backwards* from the other cases
- Observing an effect activates influences between the possible causes (two competing explanations)

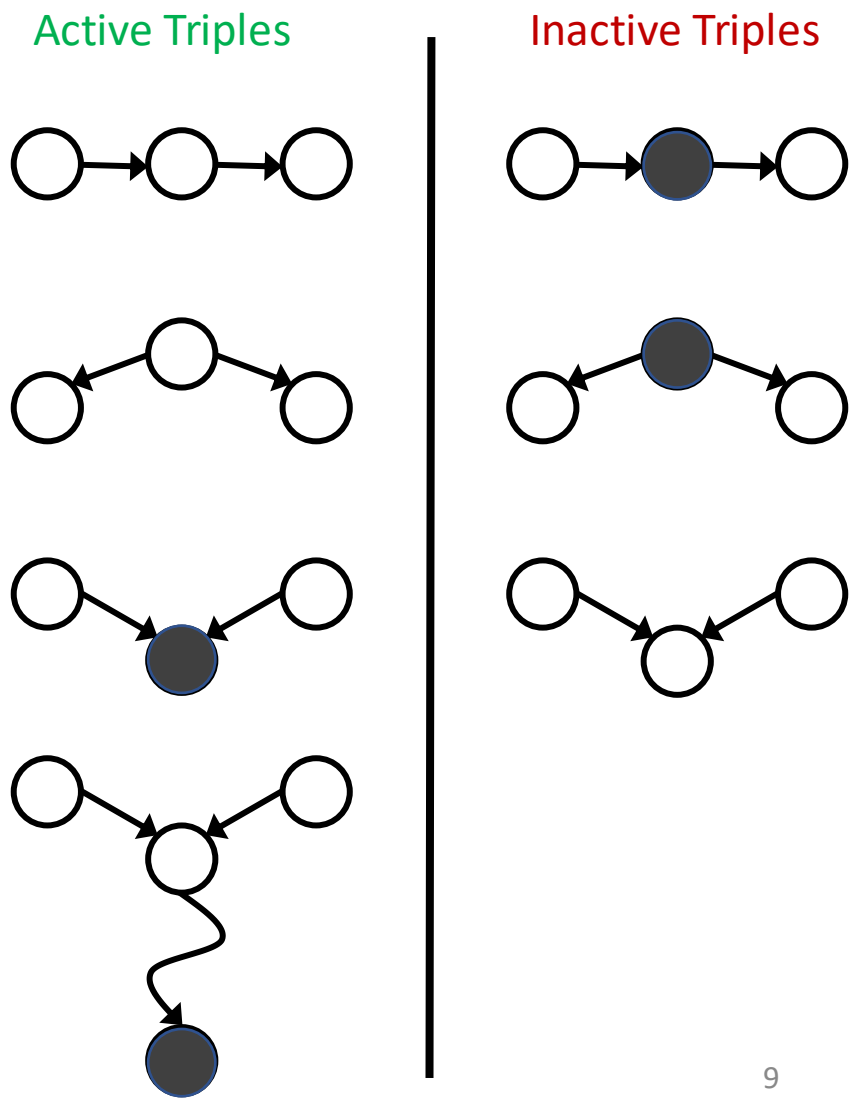
Conditional Independence

- General case: In a given BN, are two variables independent (given evidence)?
- Any complex example can be broken down into repetitions of the 3 canonical cases
- Shade the evidence nodes, look for paths in the resulting graph
- Attempt 1: If two nodes are connected by an undirected path blocked by a shaded (observed) node, they are conditionally independent
  - Works for the first two cases, but not for common effect (v-structure): the path has to be "active"

Conditional Independence

Active and inactive paths

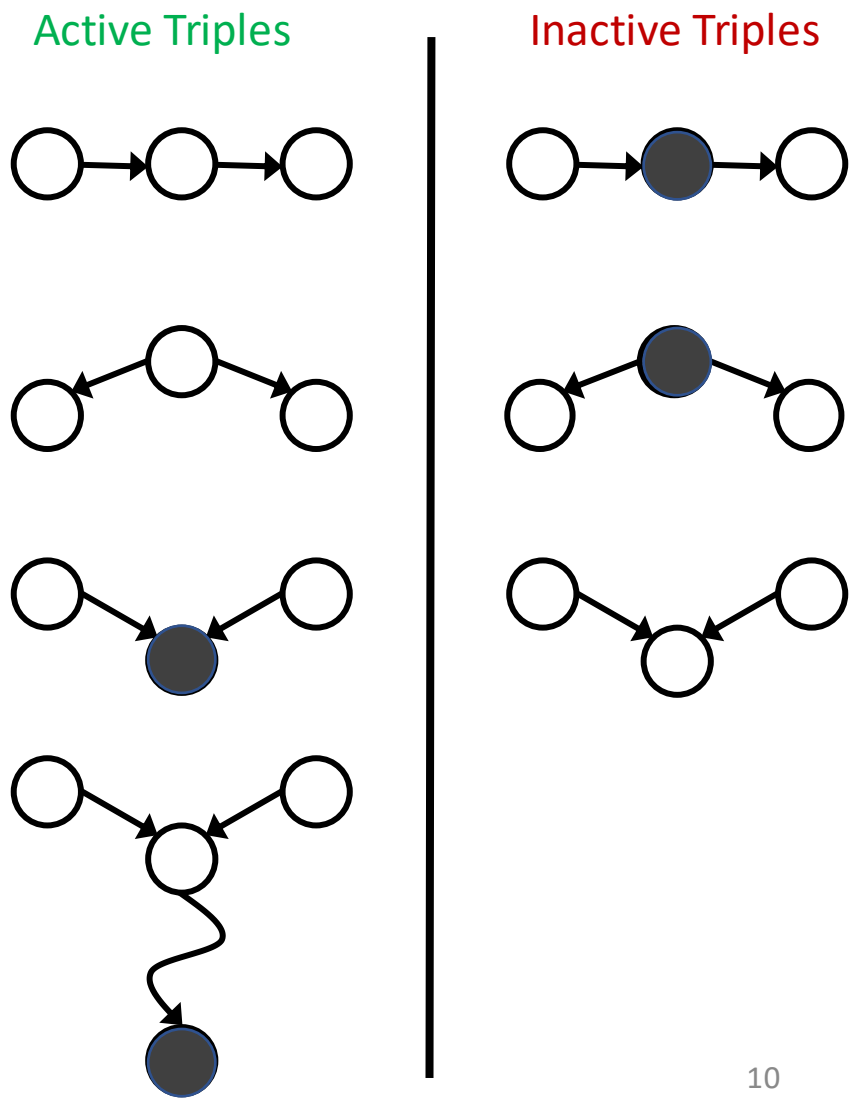
- Are X and Y conditionally independent given evidence variables Z?
- Consider all (undirected) paths from X to Y
- **No active paths = Independence**
- A path is active if **each triple** is active: *A single inactive segment blocks the path* (makes the path inactive)



Conditional Independence

A path is *active* if each triple is active:

1. **Causal chain**
   - $A \rightarrow B \rightarrow C$ where B is unobserved (either direction)
2. **Common cause**
   - $A \leftarrow B \rightarrow C$ where B is unobserved
3. **Common effect** (v-structure)
   - $A \rightarrow B \leftarrow C$ where B *or one of its descendants* is observed



Conditional Independence

D-Separation algorithm:

- Query: $X_i \perp\!\!\!\perp X_j | \{X_{k_1}, \ldots, X_{k_n}\}$
- Check all (undirected) paths between X_i and X_j
- If one or more paths are active, then independence is not guaranteed
- If **all** paths are **inactive**, then independence is guaranteed

- Given a Bayes net, can run d-separation to build a complete list of conditional independences
  - This list determines the set of probability distributions that can be represented

Conditional Independence

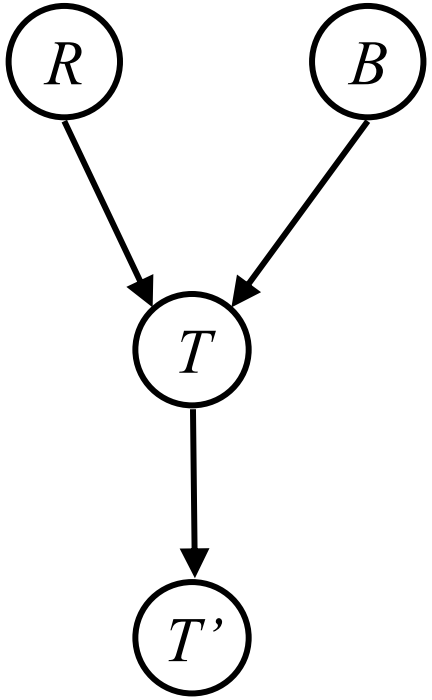
Example:

- Queries:

$R \perp\!\!\!\perp B$ *Yes*

$R \perp\!\!\!\perp B | T$ *No*

$R \perp\!\!\!\perp B | T'$ *No*



Conditional Independence

Example:

- Queries:

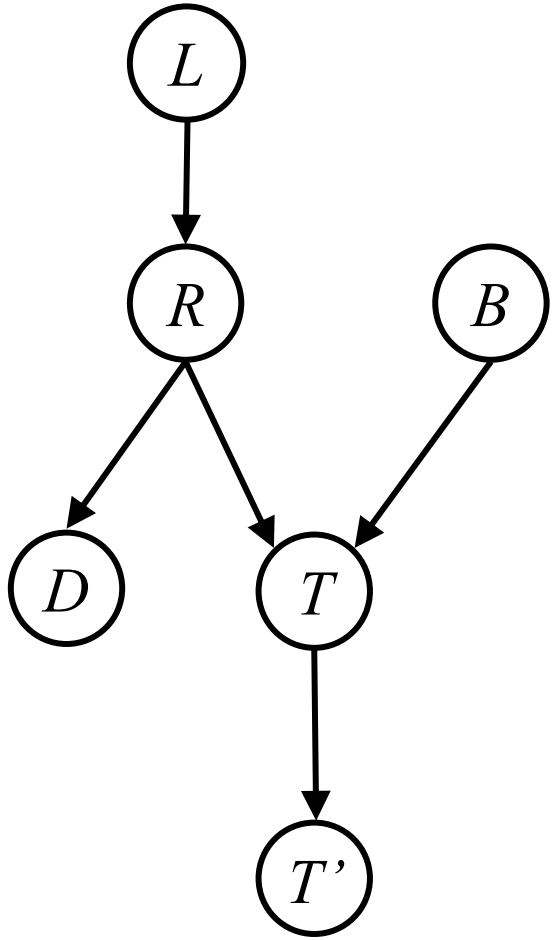
$L \perp\!\!\!\perp T' | T$ *Yes*

$L \perp\!\!\!\perp B$ *Yes*

$L \perp\!\!\!\perp B | T$ *No*

$L \perp\!\!\!\perp B | T'$ *No*

$L \perp\!\!\!\perp B | T, R$ *Yes*



Conditional Independence

Example:

- Variables:
  - R: Raining
  - T: Traffic
  - D: Roof drips
  - S: I'm sad
- Queries:

$T \perp\!\!\!\perp D$ *No*

$T \perp\!\!\!\perp D | R$ *Yes*

$T \perp\!\!\!\perp D | R, S$ *No*

